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ENGINEERING



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Saveetha Institute of Medical And Technical Sciences  
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# Multi-Agent Systems for **Disaster** Management.

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Decentralized AI coordination for real-time crisis detection,  
dynamic resource allocation, and automated response  
protocols.

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# The Current Challenge

**Data Silos:** Centralized databases create severe informational bottlenecks during high-stress, rapidly unfolding events.

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**High Latency:** Human-in-the-loop decision-making delays critical dispatch times, costing crucial minutes.

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**Static Protocols:** Legacy response plans cannot dynamically adapt to multi-front disaster scenarios.

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**Vulnerability:** Centralized command infrastructures are susceptible to localized physical damage.



# Core Architecture Roles



## SENSOR AGENTS

Deployed extensively at the edge. They gather real-time telemetry including seismic, thermal, and meteorological data, performing localized anomaly detection before transmission.



## DECISION AGENTS

Equipped with predictive models to analyze incoming verified data. They assess risk severity, calculate probabilities, and formulate optimal, bias-free resource allocation strategies.



## COMM. AGENTS

Acting as the connective tissue, these agents manage message queuing, ensure delivery protocols, and dynamically route critical packets around damaged physical infrastructure.

# Agent Coordination Strategies

## Decentralized Consensus

In highly volatile zones, agent clusters negotiate immediate resource allocation without central oversight. They utilize auction-based algorithms to rapidly prioritize tasks based on threat proximity.

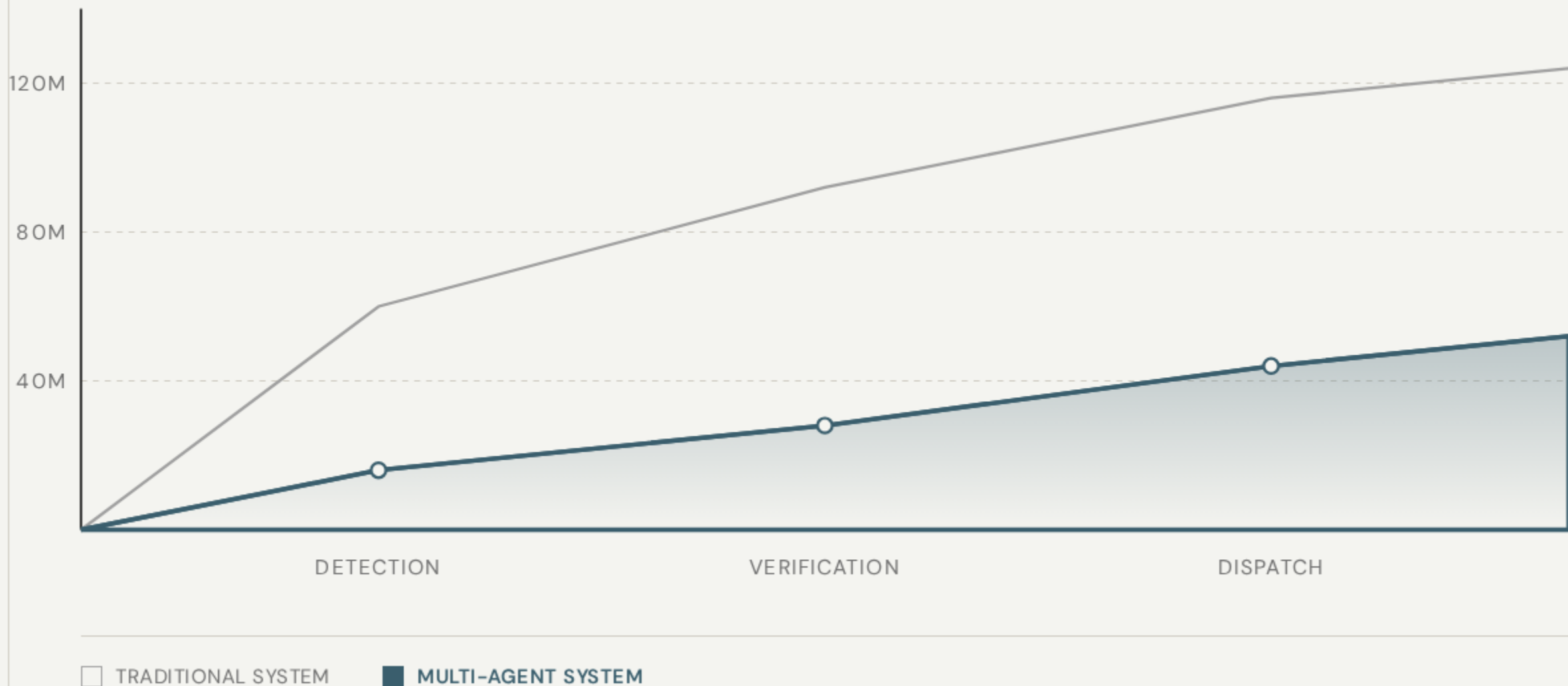
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## Hierarchical Reporting

For macro-level strategic planning, lower-level agents aggregate and summarize granular data. They pass only high-confidence state assessments upward, filtering out noise for regional responses.

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# Response Latency Over Crisis Stages



# Paradigm Shift: Legacy vs. MAS

| System Characteristic | Traditional Architecture      | Multi-Agent System        |
|-----------------------|-------------------------------|---------------------------|
| Decision Making       | Centralized Command Center    | Distributed & Autonomous  |
| Fault Tolerance       | Low (Single point of failure) | High (Self-healing mesh)  |
| Scalability           | Limited by central bandwidth  | Virtually Unlimited (P2P) |
| Adaptability          | Static / Pre-programmed       | Dynamic / ML-driven       |
| Data Processing       | Post-event batch processing   | Real-time edge streaming  |